Charles Allison

Worcester, MA | (508) 579-9494 | charlesallisonjr20@gmail.com
LinkedIn: www.linkedin.com/in/charles-allison-jr

EDUCATION

Worcester Polytechnic Institute Worcester, MA
Bachelor's Degree Candidate Expected May 2028
Biology & Biotechnology

PROJECTS

Quantitative Analysis of Ecological and Environmental Datasets Apr 2026
Portfolio Research Project

- Analyzed four ecological datasets involving wildlife crossings, island fox populations, coral bleaching, and ecosystem carbon storage.
- Organized and visualized data, identified key variables, and interpreted biological and environmental trends.
- Applied descriptive and inferential statistics (mean, standard deviation, t-tests, ANOVA) to evaluate group differences.
- Interpreted results using p-values and hypothesis testing, producing a quantitative research report on conservation, biodiversity, and climate-related impacts.

LABORATORY EXPERIENCE

Antimicrobial Resistance & Molecular Analysis — *Worcester Polytechnic Institute*

- Isolated rifampicin-resistant *E. coli* colonies using aseptic technique, serial dilution, antibiotic selection, plating, incubation, and colony counting.
- Amplified the *rpoB* hotspot region using PCR, confirmed expected DNA fragments by agarose gel electrophoresis, and purified PCR products for sequencing.
- Analyzed sequencing results in Benchling and identified a C1535A point mutation causing an S512Y amino acid substitution associated with rifampicin resistance.

DNA Purification & Restriction Mapping — *Worcester Polytechnic Institute*

- Purified plasmid DNA from bacterial culture using alkaline lysis and silica spin column purification.
- Quantified plasmid DNA using NanoDrop spectrophotometry, recording **36.4 ng/μL** and calculating a total yield of **1.82 μg**.
- Performed restriction enzyme digestion and agarose gel electrophoresis to compare DNA fragment patterns and interpret plasmid identity.

Protein Quantification & Microbial Growth Analysis — *Worcester Polytechnic Institute*

- Measured enzyme activity using spectrophotometry and analyzed absorbance changes across temperature and pH conditions.
- Used protein assay data and a standard curve to calculate unknown protein concentration and compare assay results.
- Practiced aseptic technique, bacterial culture, serial dilution, viable cell counts, OD measurements, CFU/mL calculations, and antibiotic growth comparison using *E. coli*.

PROFESSIONAL EXPERIENCE

Worcester Polytechnic Institute Worcester, MA
Marketing Assistant May 2025 – Present

- Create and edit Instagram reels, videos, and carousel posts for WPI events, resources, and campus life.
- Increased profile views, reach, and engagement analytics by over 15%.
- Use engagement metrics and informal student feedback to evaluate what content keeps students engaged, what causes viewers to click away, and which topics perform best.

SKILLS

Lab: Micropipetting, aseptic technique, bacterial culture, serial dilution, plating, PCR, gel electrophoresis, restriction digest, plasmid purification, NanoDrop, spectrophotometry, protein assays, Benchling
Digital: Microsoft Office, Google Workspace, Canva, Instagram analytics, video editing

ADDITIONAL

Certifications & Training: BLS for Healthcare Providers Certified, OSHA Certified, Diesel Technology Certified
Awards: Freshman of the Year — WPI (2025), Certificate of Recognition — Mayor Joseph M. Petty (2024)
Interests: Photography, Hiking, Running, & Bodybuilding